

CSA0707

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SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct: (192611088)

I GEORGE ALLIS J (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: George

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Q3 10 marks

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

QUESTION - 1 [SUBNET DEBUGGING]

1) Problem identification:

Given IP : 192.168.10.65/26

- Subnet : 192.168.10.64/26
- Network Address : 192.168.10.64
- Broadcast Address : 192.168.10.127
- Valid Host Range : 192.168.10.65 - 192.168.10.126

2) Debugging

→ IP 192.168.10.65 is valid.

→ Gateway 192.168.10.1 is invalid

because it belongs to 192.168.10.0/26.

→ Therefore, the workstation cannot properly communicate with the intended subnet.

3) Optimization:

Use:

- ▲ IP Address : 192.168.10.66
- ▲ Subnet Mask : 255.255.255.192
- ▲ Default Gateway : 192.168.10.65

4) Analysis :

Both IP and gateway must belong to the same subnet for correct local communication.

5) Result :

$$\begin{aligned}\text{Usable hosts} &= 2^6 - 2 \\ &= 62\end{aligned}$$

Final : corrected subnet is 192.168.10.0/26
with 62 usable host address.

QUESTION : 2 [IPV6 DEBUGGING]

Given :

- Device A : 2001 : db8 :: f00 : 42 : 8329 / 64
- Device B : 2001 : db8 :: 0 : 0 : 1 : : 1 / 64

1) Problem identification :

Expanded addresses

→ A : 2001 : 0db8 : 0000 : 0000 : ff00 : 0042 : 8329

expansion is 2001 : 0db8 : 0000 : 0000 : ff00 : 0042 :
8329

→ B 2001 : db8 : 0000 : 0001 : 000 : 000 : 0001

2) Debugging

- Device A prefix : 2001 : db8 : 0 : 0 : : /64
- Device B prefix : 2001 : db8 : db8 : 0 : 0 : 1 : : /64
- They are not in the same /64 subnet

3) Optimisation :

- Device A prefix : 2001 : db8 : 0 : 0 : : /64
- Device B prefix : 2001 : db8 : 0 : 0 : 1 : : /64
- They are not in the same /64 subnet

4) Analysis :

Both devices now share the same /64 network prefix, so communication is possible

5) Result :

A /64 IPv6 subnet contains 2^{64} addresses.

3) QUESTION [IP V4 SUBNET ALLOCATION & VLSM]

Problem identification:

- $/26$ provides 62 hosts
- $/27$ provides 30 hosts
- $/28$ provides 14 hosts
- Giving the same size subnet to every department may waste address.

Debugging:

check whether the required number of hosts matches the allocated subnet. A department needing only hosts does not require a $/26$.

Optimise - VLSM

Use different subnet sizes according to department requirements.

- Large department $\rightarrow /26$
- Medium department $\rightarrow /27$
- Small department $\rightarrow /28$

Analysis:

VLSM reduces unused IP addresses and improves overall address utilization.

Conclusion: VLSM is more efficient than fixed-size subnetting.

QUESTION: 4 (IPV6 DEBUGGING)

Problem:

A and B have different /64 network prefixes.

Debugging:

- A prefix : $2001 : db8 : 0 : 0 :: /64$
- B prefix : $2001 : db8 : 0 : 0 : 1 :: /64$
- Different subnet $\rightarrow$ communication fails.

Optimization:

$\Rightarrow$ Assign both to same subnet.

$\Rightarrow$ A : $2001 : db8 : 0 : 0 : 1 :: 10 /64$

$\Rightarrow$ B : $2001 : db8 : 0 : 0 : 1 :: 20 /64$

Result :

Both devices are now in the same /64

subnet:

Available IPv6 addresses = 2^{64}

QUESTION: 5:

1) Destination: 192.168.2.100

Problem identification

192.168.2.100 belongs to 192.168.2.0/24.

2) Debugging:

Longest prefix match

192.168.2.0/24 $\rightarrow$ 10.0.0.3

Therefore, the selected next hop is:

10.3.0.0.3

3) Optimization:

keep the specific /24 route too
because it provides a more precise

path than the default route

Analysis :

The router selects the most specific matching route :